

RISHIKESH PATIL

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SUMMARY

Final-year B.Tech student in Electronics & Computer Science (**CGPA: 9.65/10, Ranked Top 5 in Department**) with hands-on experience across Applied AI/GenAI engineering (RAG pipelines, agent orchestration), NLP research, and full-stack development. Currently an SDE Intern at AyriTech; previously built production software at Tata Consulting Engineers. Author on research spanning speech emotion recognition and continual learning, with an original SER framework (ACE-SER) currently filed for institutional copyright. Led a 20+ member IEEE technical team. Seeking Applied AI/GenAI Engineer roles focused on RAG and agent orchestration.

EDUCATION

Vidyalankar Institute of Technology

B.Tech in Electronics and Computer Science (Final Year); **CGPA: 9.65/10 | Rank: Top 5 in Department**

Mumbai, Maharashtra

Aug 2023 – Present

DAV School

10th Board (S.S.C): 89.4% | 12th Board (H.S.C): 88%

Mumbai, Maharashtra

2023

RESEARCH PAPERS (In Preparation – Targeting Submission 2026)

ACE-SER: Adaptive, Efficient & Explainable Speech Emotion Recognition | Targeting: IEEE/Interspeech/ICASSP

2026

- Final Year Project introducing three novel modules – **ASCA** (Adaptive Speech Complexity Analyzer, scoring acoustic complexity from spectral entropy, pitch variance, RMS energy, and HNR), **GIE** (Green Inference Engine routing samples between lightweight MFCC and heavy SSL-backbone pipelines by complexity score), and **ASRE** (Audible Saliency Reconstruction Engine, producing audible Grad-CAM explanations via phase-preserving inverse STFT).
- Institutional **copyright application filed** (in process) for the ACE-SER architecture.

Comprehensive Survey on Speech Emotion Recognition | Targeting: Springer Journal

2026

- PRISMA-based review of **60 papers** across traditional ML, deep learning, hybrid/multimodal, SSL (wav2vec 2.0, HuBERT, WavLM), and multilingual SER paradigms.
- Identifies open gaps in cross-corpus generalization, class imbalance, explainability (XAI/Grad-CAM), and Green AI deployment for real-world HCI systems.

Adaptive Fisher Regularization for Continual Learning in ANFIS | Conference

2026

- Proposed the **Rule Drift Index (RDI)**, a novel metric quantifying structural drift in fuzzy membership functions during sequential neuro-fuzzy learning, evaluated on wine quality and building energy datasets.
- Empirically found sequential learning outperformed all adaptive/fixed EWC variants on the energy dataset with statistical significance – a result reported transparently to inform when regularization-based continual learning is (and isn't) warranted.

PROFESSIONAL EXPERIENCE

SDE Intern

AyriTech

May 2026 – Present

Hybrid

- Contributing to a multi-modal emotional-AI agent product within a 20+ engineer cohort, building RAG-based conversational memory (LangChain, vector database retrieval) and async backend routing (FastAPI, WebSockets, Redis session state).
- Working across an agile, sprint-based delivery cycle involving audio/text emotion inference pipelines and real-time LLM-driven response generation, under mentorship from senior engineering leads.

Software Development Intern

Tata Consulting Engineers Limited (TCE)

Dec 2025 – Jan 2026

Mumbai, India

- Built **SmartCafe**, a full-stack Cafe Management System (React, PostgreSQL) with inventory tracking, order processing, and analytics dashboards for enterprise workflows.

Technical Head – IEEE Student Branch

Vidyalankar Institute of Technology

Sep 2025 – Present

Mumbai, India

- Lead **20+ developers**; organize technical workshops, hackathons, and full-stack/ML mentorship programs; spearheaded migration of the [IEEE VIT Mumbai website](#) to React with Supabase backend ([GitHub](#)).
- Drove **~35%** increase in event engagement through interactive UI, dynamic event pages, and Git-based agile collaboration across the team.

PROJECTS

FinGuard AI | Multi-Agent Finance Fraud Detection – LangGraph, RAG, FastAPI, React (In Progress)

2026

- Building a multi-agent fraud detection system with RAG pipelines and LangGraph orchestration, LangSmith tracing, RAGAS-based evaluation, and LoRA/QLoRA fine-tuning, served via FastAPI with a React frontend.

Faculty Evaluation App | React.js, Vite, Supabase (PostgreSQL), CSS, Netlify

[GitHub](#) | Jun – Sep 2025

- Web application to digitize faculty feedback with React/Vite frontend, Supabase for real-time database and auth, and admin dashboards for insights; deployed on Netlify under principal's mentorship.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, HTML/CSS
AI/ML & GenAI: LangChain, LangGraph, RAG, Vector Databases, LoRA/QLoRA, PyTorch, TensorFlow, Scikit-learn, BERT/Transformers, wav2vec 2.0, ANFIS, SVM
Frameworks & Tools: React.js, Vite, FastAPI, Flask, Git & GitHub, Netlify, Postman
Databases: PostgreSQL, Supabase, SQL
Core Competencies: Agent Orchestration, ML Research, Full-Stack Development, RESTful APIs, Continual Learning, Speech & Legal NLP

ACHIEVEMENTS & ADDITIONAL INFORMATION

CDAC – Cloud Computing (Short-Term Course) *Centre for Development of Advanced Computing*
• Trained on cloud fundamentals, Linux, and virtualization; hands-on with cloud deployment workflows.
Certifications: MATLAB Onramp & Control System Design using Simulink (MathWorks) | TCSion Soft Skills
Languages Spoken: English, German, Hindi, Marathi